International and Monetary Economics

Course Code: 30054

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Lecture 5

Roadmap

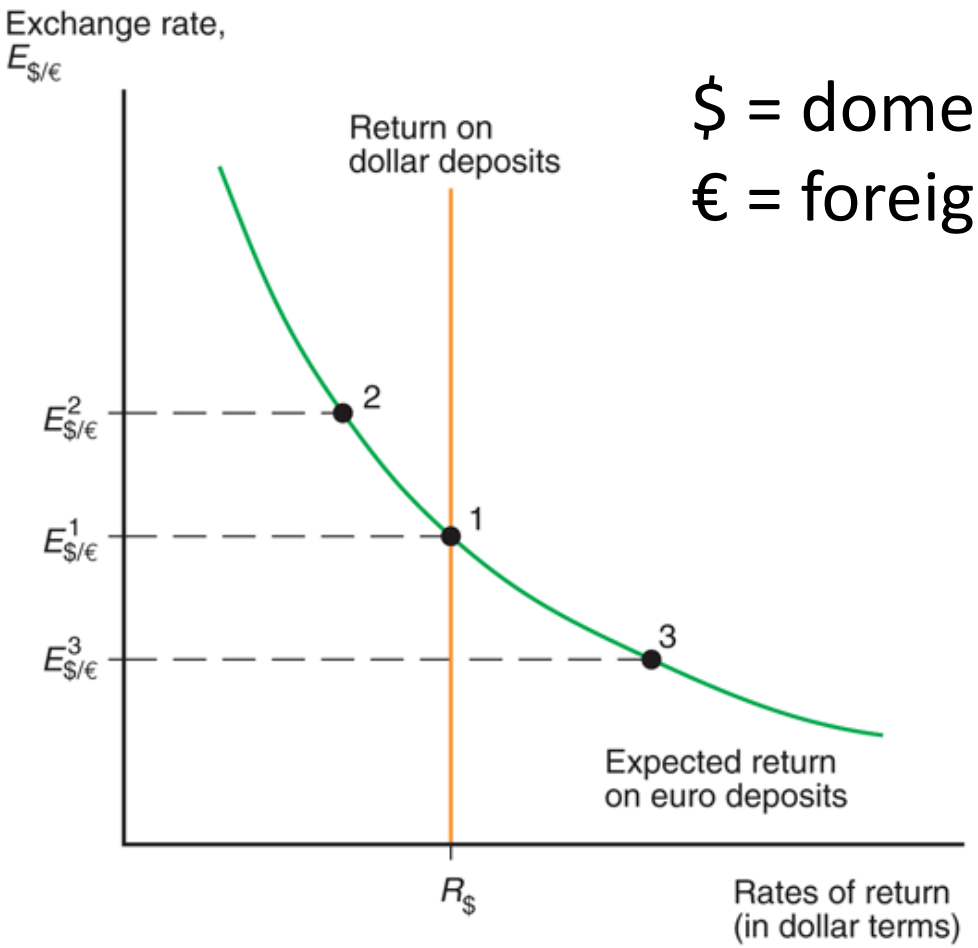
1. Interest rate parity
   - A useful diagram
   - UIP vs. CIP
   - Empirical evidence

2. UIP and the Carry Trade

3. Money
   - Supply
   - Demand

A Useful Diagram

A Convenient Way to Represent the Parity



$ = domestic currency

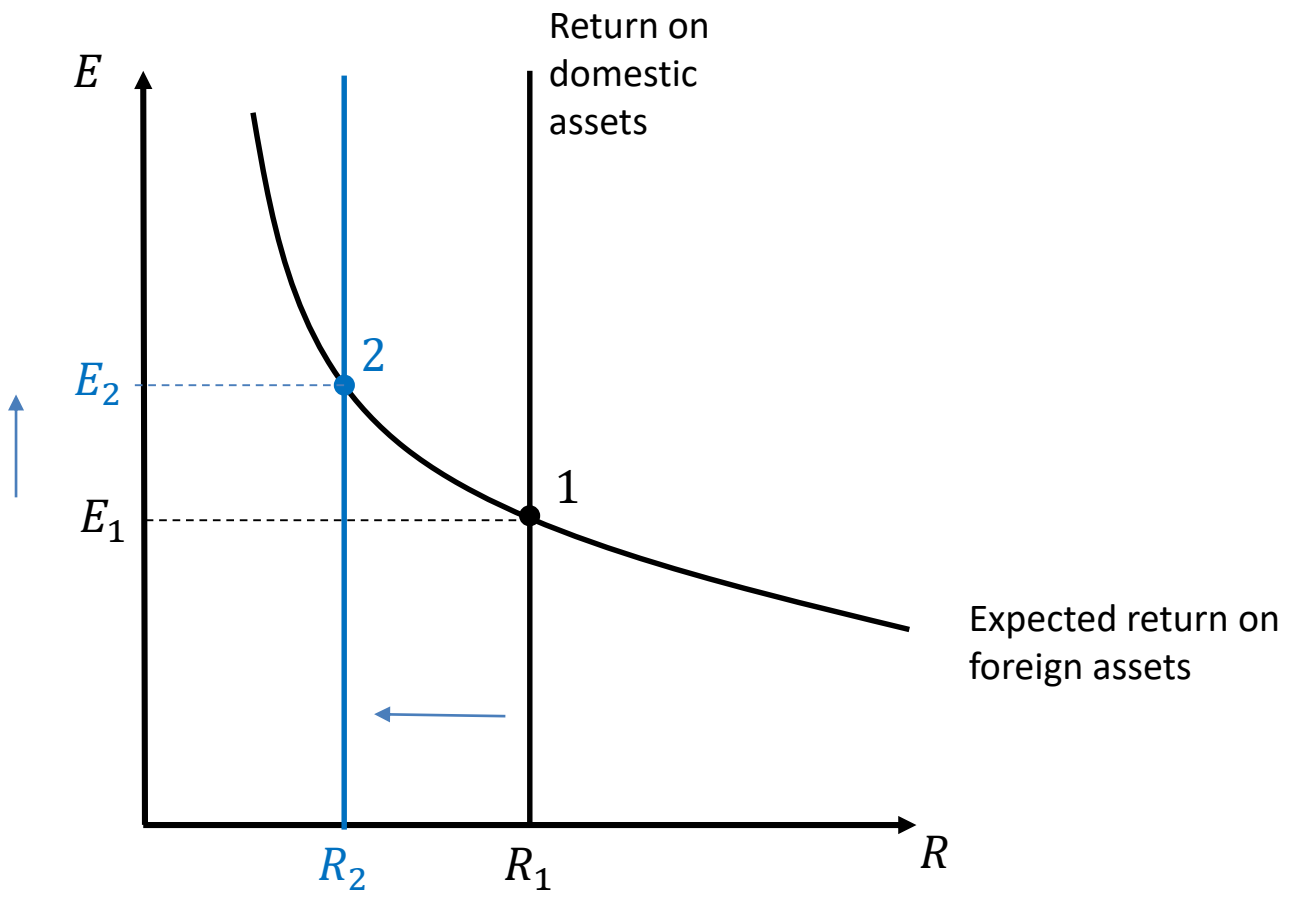
€ = foreign currency

$$R_\$ = R_€ + \frac{E^e - E}{E}$$

Note: $R_€$ and E^e are fixed on this diagram

Exercise 1

A Decrease in Domestic Interest Rate



At E_1, the expected rate of return on foreign assets is higher than R_2

$\Downarrow$

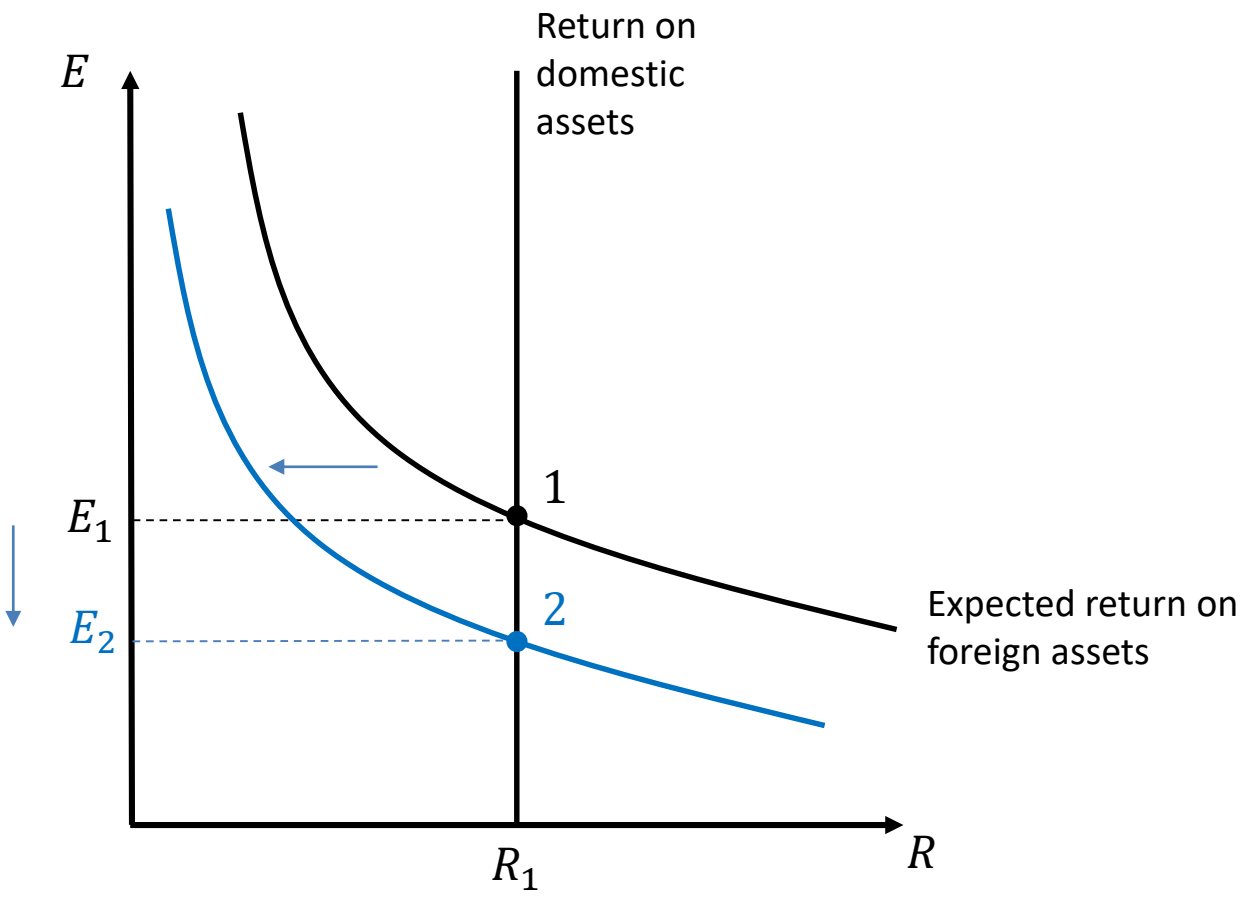
higher demand for foreign currency

$\Downarrow$

the domestic currency depreciates

Exercise 2

A decrease in the foreign interest rate



At E_1, the expected rate of return on foreign assets is lower than R_1

$\Downarrow$

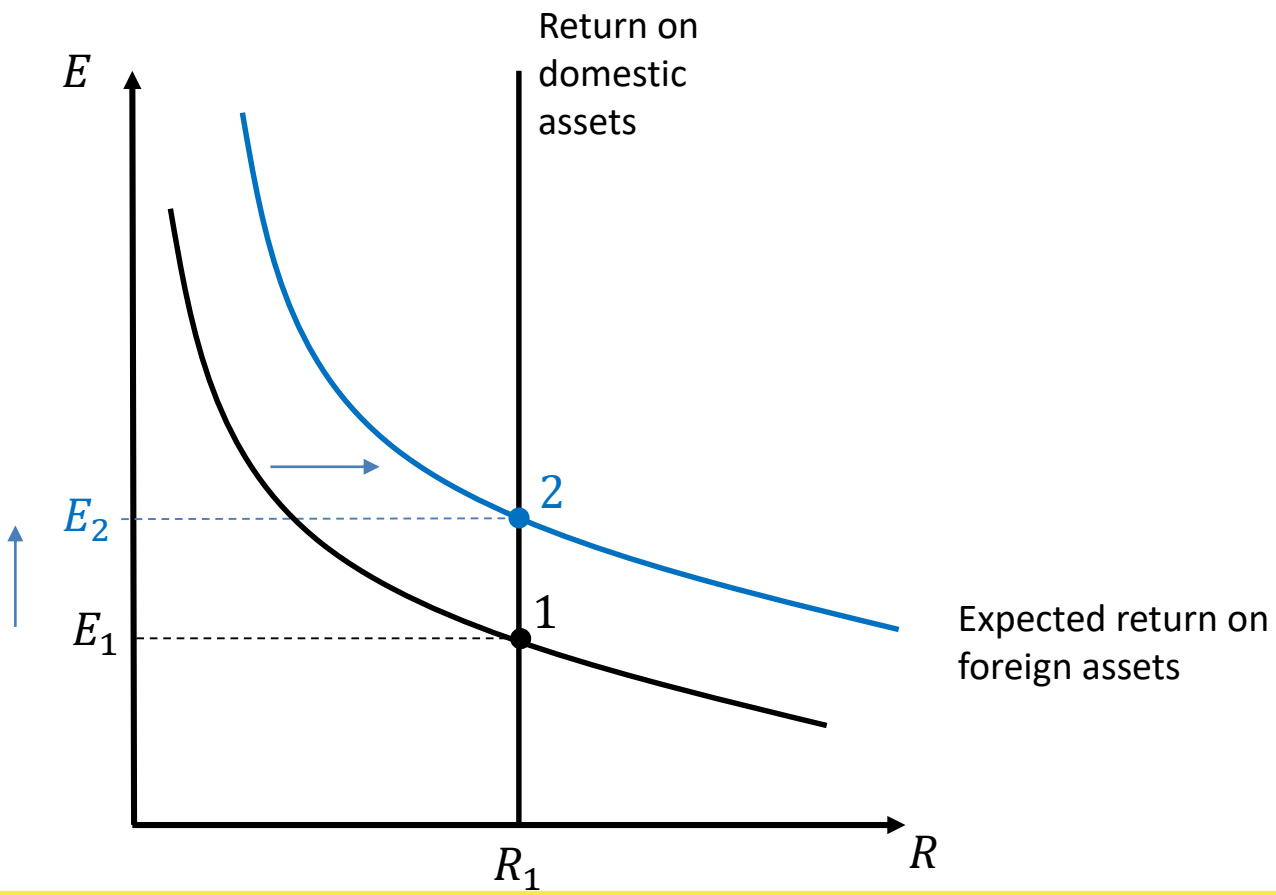
higher demand for domestic currency

$\Downarrow$

the domestic currency appreciates.

Exercise 3

An increase in E^e



$E^e \uparrow$

$\Downarrow$

higher demand for foreign currency

$\Downarrow$

the foreign currency appreciates/the domestic currency depreciates

The exchange rate behaves like any other asset price: its value today depends on expectations of its value in the future

Important Note

- Interest rate parity

$$R = R^* + \frac{E^e - E}{E}$$

- Appreciate the difference
  - If R^* is fixed, an increase in R increases the ***rate*** of depreciation of home currency
  - If, in addition E^e is fixed, we can conclude that an increase in R appreciates home currency on impact

UIP vs CIP

Interest Parities

1. Covered interest parity (CIP)

$$R = R^* + \frac{F - E}{E}$$

   - F is the forward exchange rate
   - Forward exchange contract covers the FX risk

2. Uncovered interest parity (UIP)

$$R = R^* + \frac{E^e - E}{E}$$

   - Uncovered because exposes to exchange rate risk

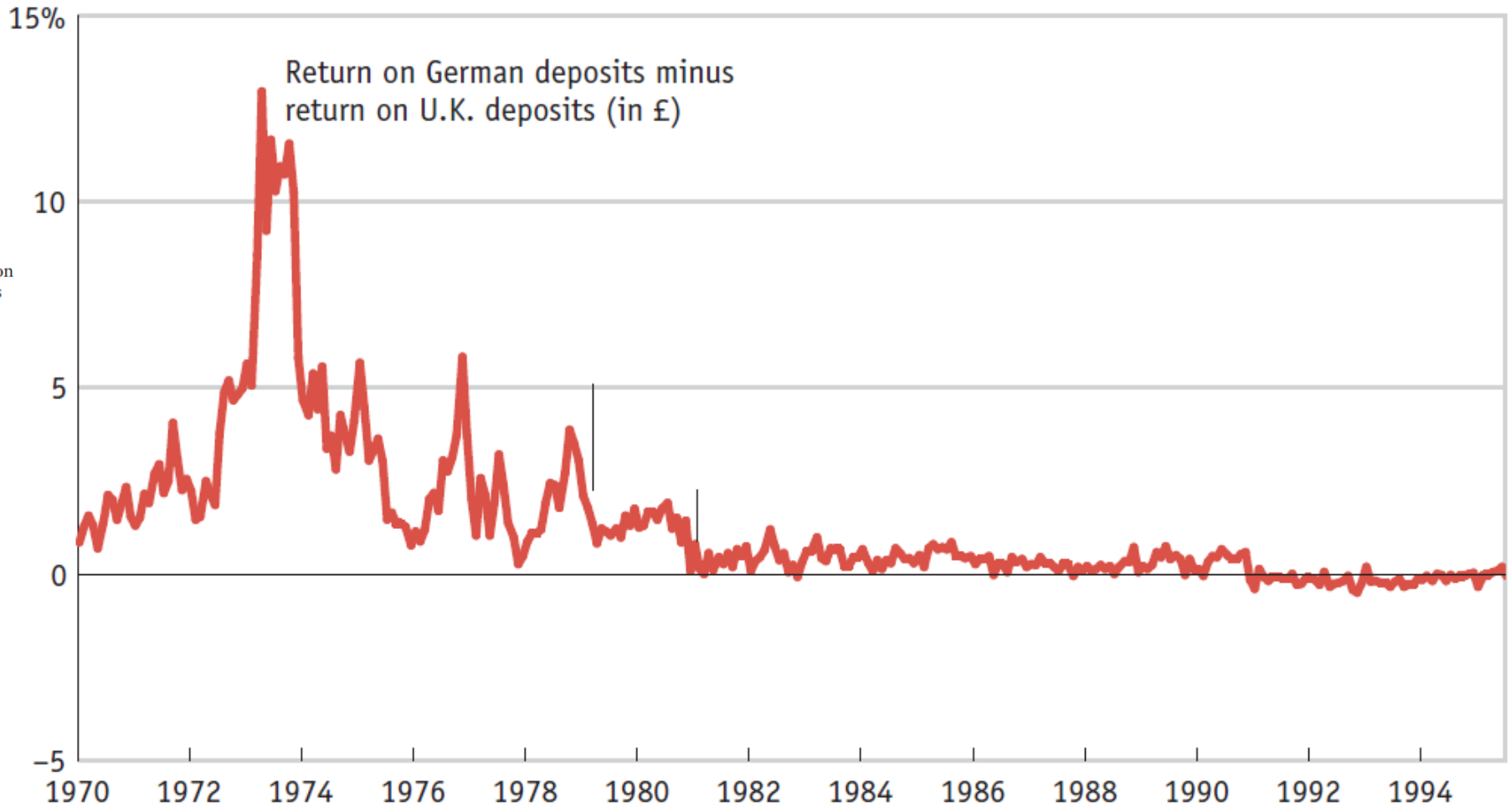
Empirical Evidence

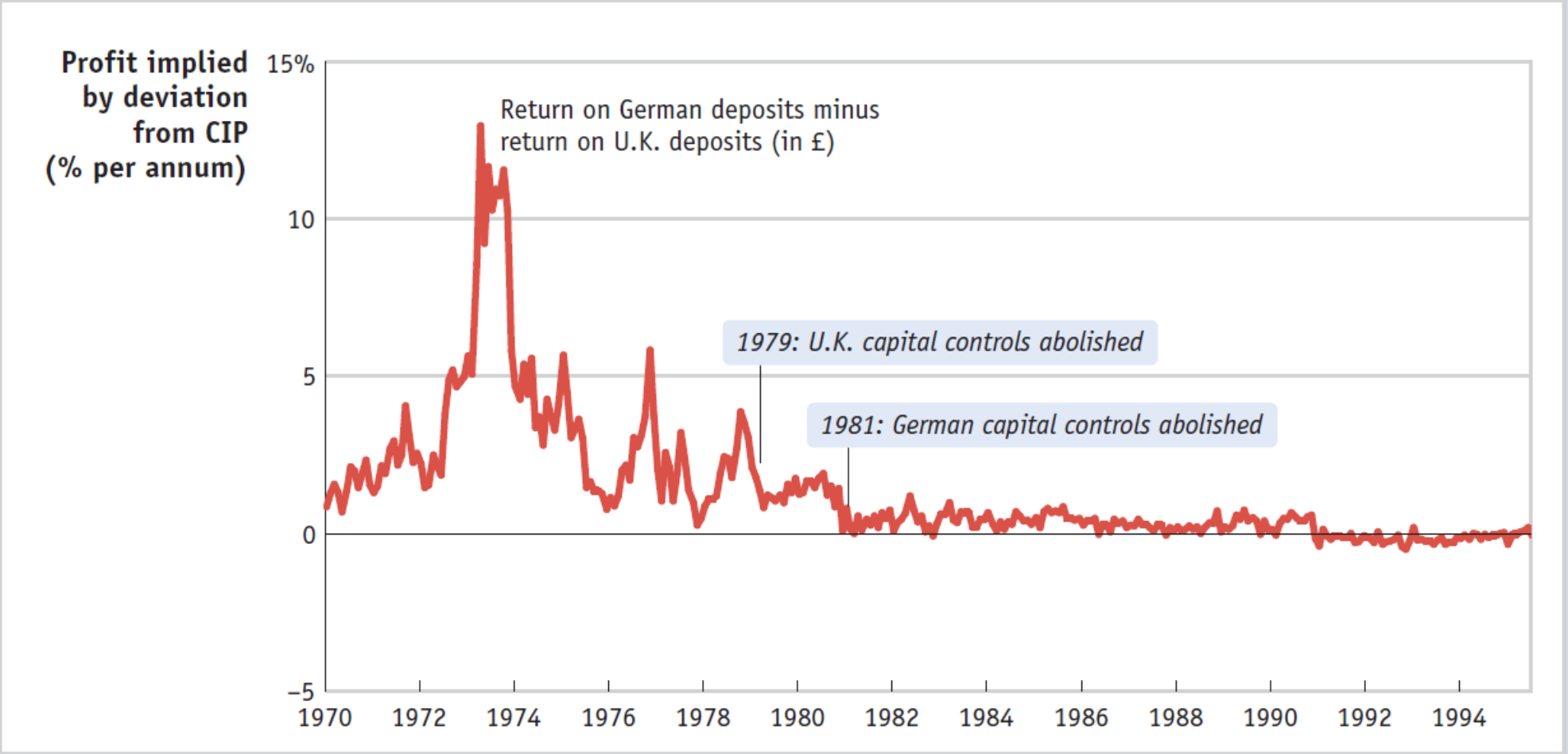
on CIP and UIP

Evidence on CIP

Profit implied by deviation from CIP (% per annum)

$$\text{Profit} = \underbrace{(1 + i_{GER})\frac{F_{UK/GER}}{E_{UK/GER}}}_{\text{Pound return on German deposits}} - \underbrace{(1 + i_{UK})}_{\text{Pound return on U.K. deposits}}$$



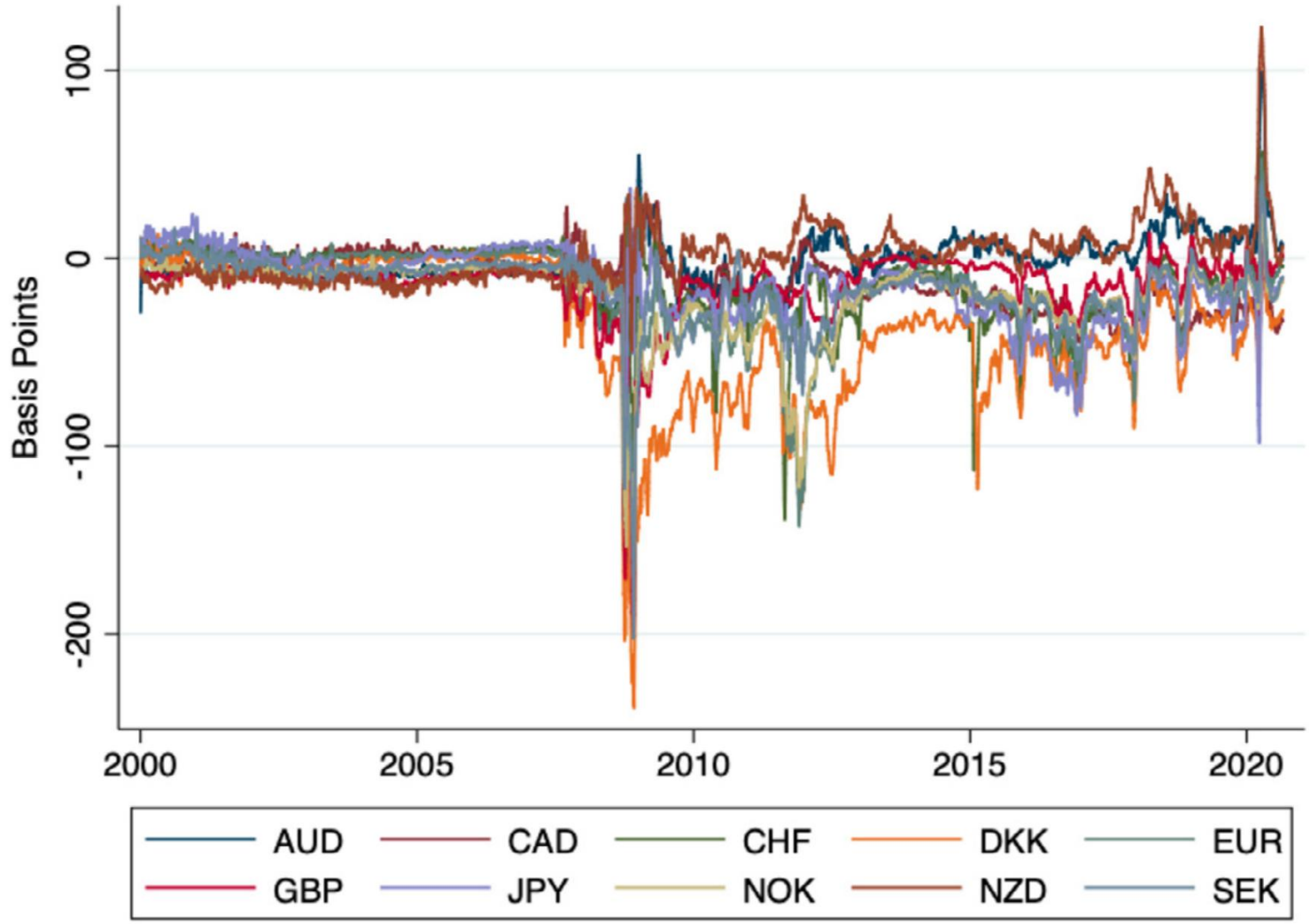


Financial Liberalization and Covered Interest Parity: Arbitrage between the United Kingdom and Germany The chart shows the difference in monthly pound returns on deposits in British pounds and German marks using forward cover from 1970 to 1995. In the 1970s, the difference was positive and often large: traders would have profited from arbitrage by moving money from pound deposits to mark deposits, but capital controls prevented them from freely doing so. After financial liberalization, these profits essentially vanished, and no arbitrage opportunities remained. The CIP condition held, aside from small deviations resulting from transactions costs and measurement errors.

Source: Maurice Obstfeld and Alan M. Taylor, 2004, Global Capital Markets: Integration, Crisis, and Growth, *Japan–U.S. Center Sanwa Monographs on International Financial Markets (Cambridge, UK: Cambridge University Press).*

CIP during and after the Global Financial Crisis

(A) Three-Month Deviations

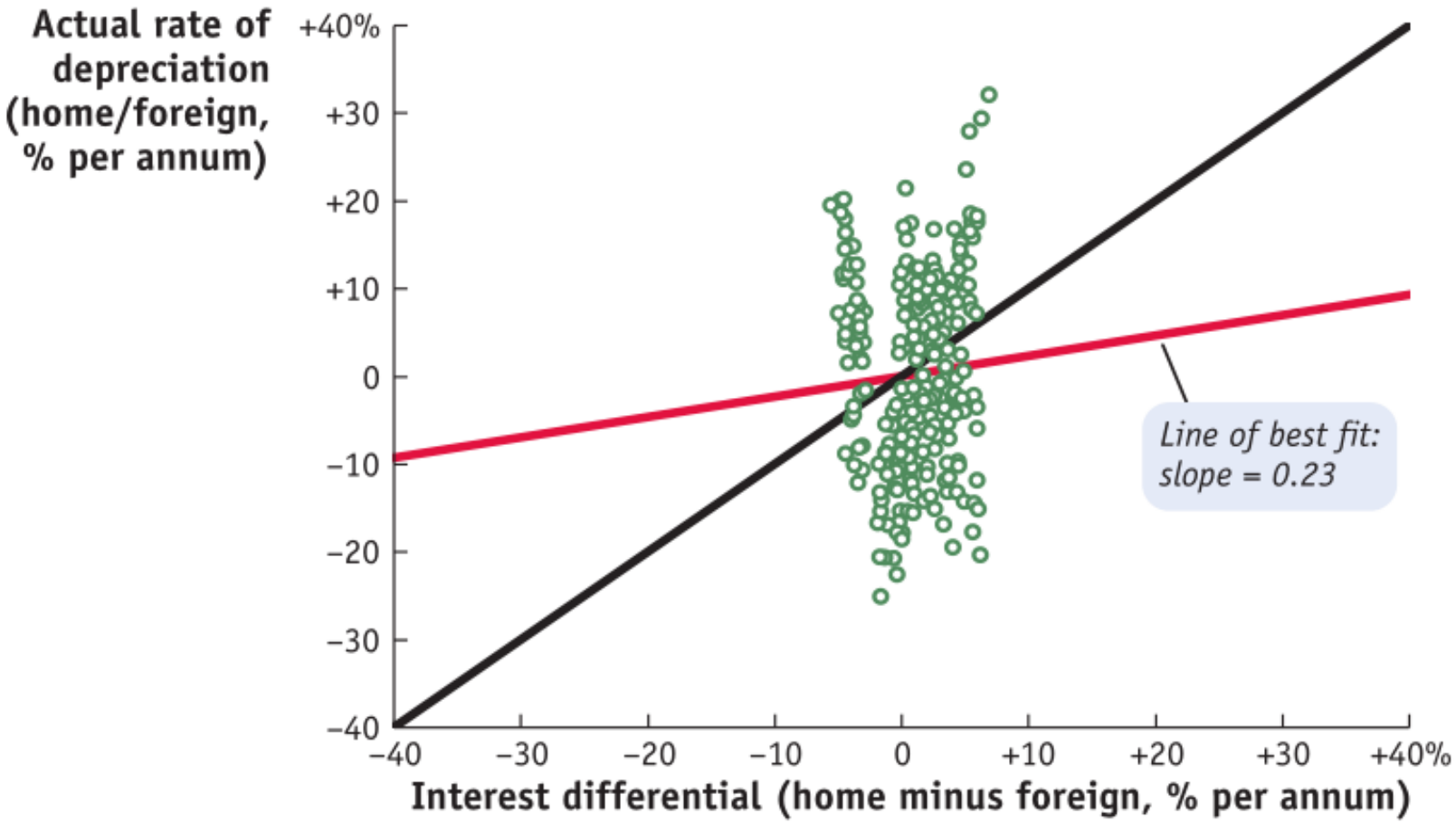


Source: Du, Schreger (2022)

Evidence on CIP

- CIP holds remarkably well *most* of the time
- But sometimes it does not hold because of
  - capital controls
  - counterparty/liquidity risk
  - Financial regulations
- Examples of the recent deviations
  - Following the collapse of Lehman Brothers, counterparty risk increases enormously
  - Persistent deviations after that due to new regulations that limit financial positions of banks

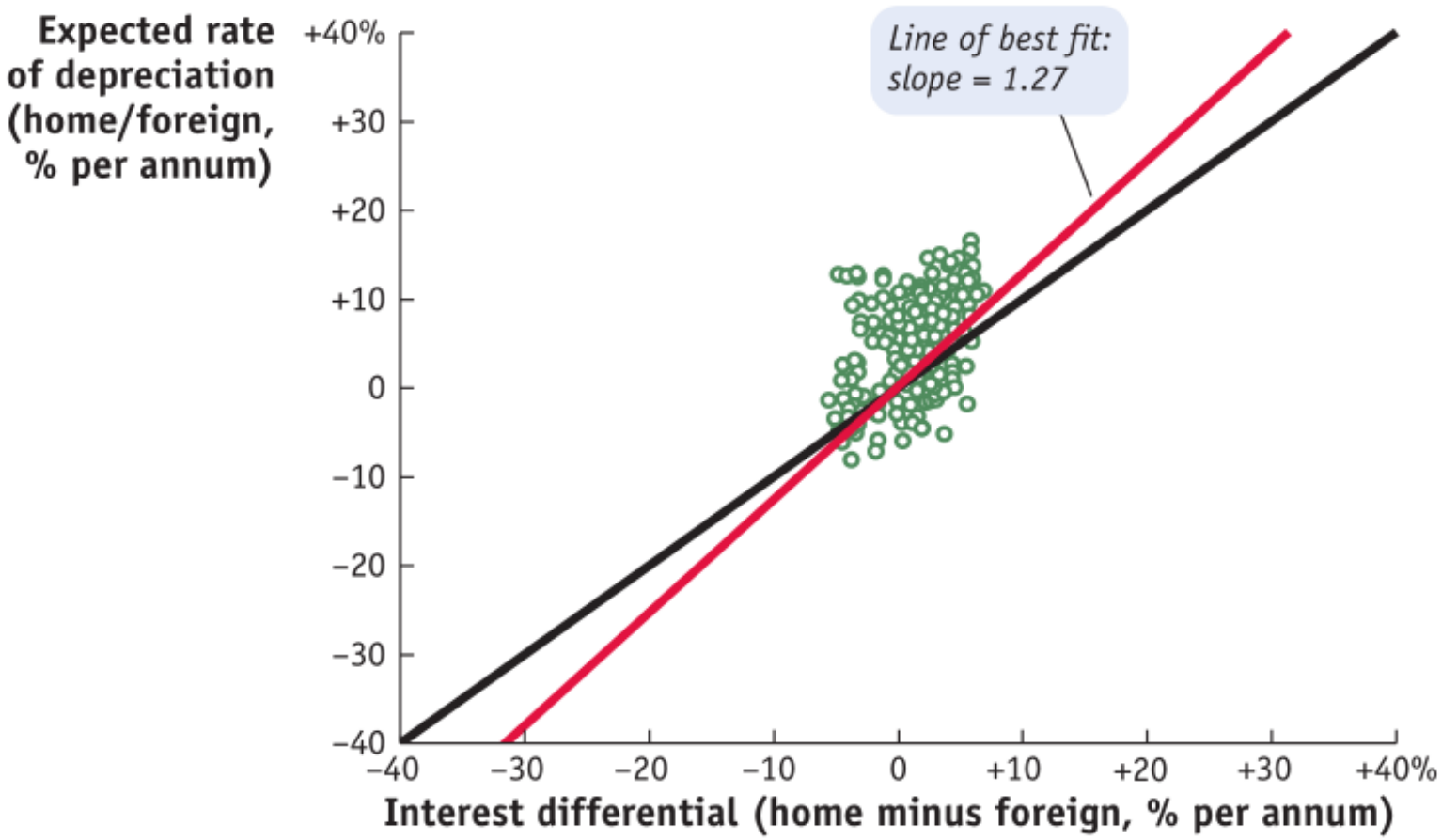
Evidence on UIP



- A **realized (actual)** rate of depreciation on Y axis
- The plot shows that
  - UIP does not seem to hold
  - This observed failure of the UIP is usually called **the forward premium puzzle**

Interpreting the slope of the best fit line: for every 1% of interest differential in favor of the domestic currency, we would expect only a 0.2% increase in **the rate** of depreciation of the home currency, on average, leaving a profit of 0.8%.

Evidence on UIP



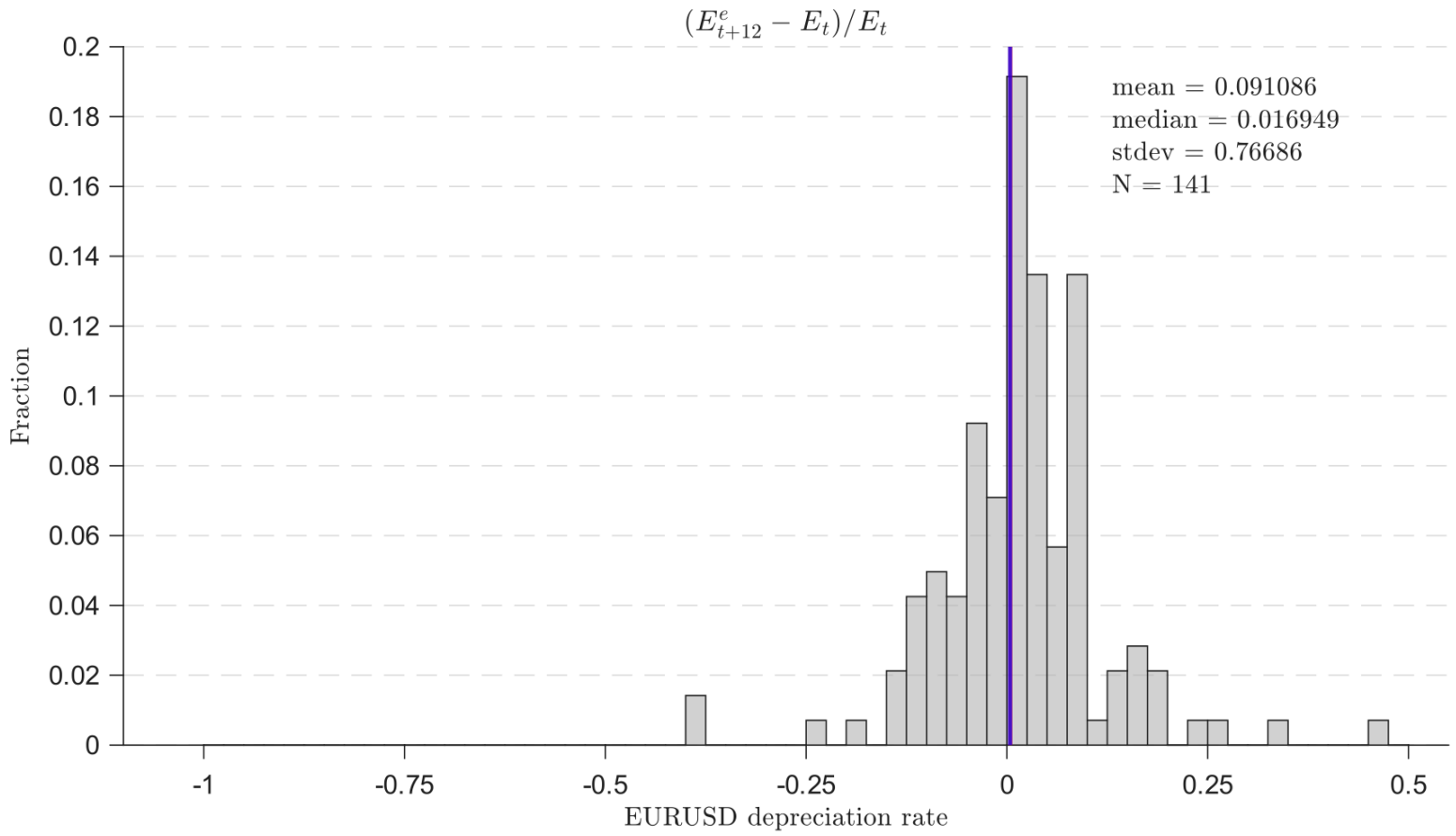
- An **expected** rate of depreciation on Y axis (surveys of currency traders)
- The plot shows that
  - UIP does not seem to hold **precisely**
  - However, there is a good positive relation

- **Based on this evidence, we will treat the UIP as a useful approximation**

Notes:
- Data are monthly for the German mark, Swiss franc, Japanese yen, British pound, and Canadian dollar against the U.S. dollar from February 1988 to October 1993

Source:
- Menzie Chinn and Jeffrey A. Frankel, 2002, "Survey Data on Exchange Rate Expectations: More Currencies, More Horizons, More Tests."

Lecture 1 Survey (Feb 5, 2026 + Feb 9, 2026)



Market data on **Feb 5, 2026**

- $R^{US} - R^{EUR} = 0.004$
- $\frac{F-E}{E} = 0.0049$

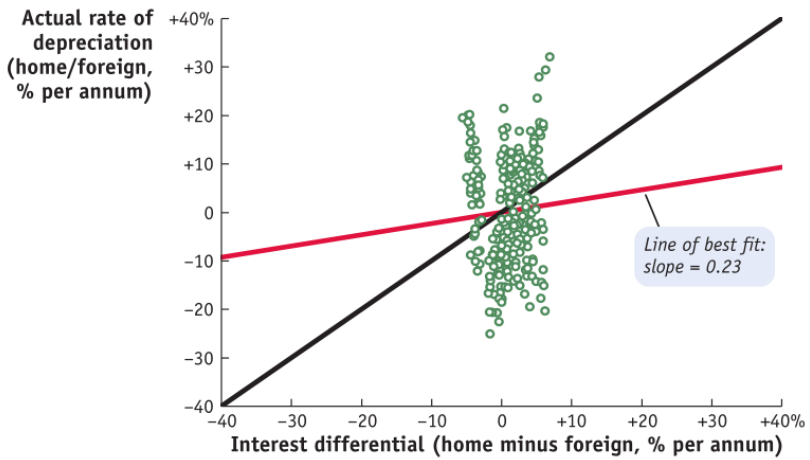
Data sources:

- Historical FX: Barchart
- Historical interest rates: FRED

UIP and the Carry Trade

Carry Trade

- Motivation



- A carry trade investment strategy
  - **Borrow** in a low interest rate currency (funding currency)
  - **Invest** in a high interest rate currency (target currency)
  - **Pray** that the funding currency doesn't appreciate against the target currency!

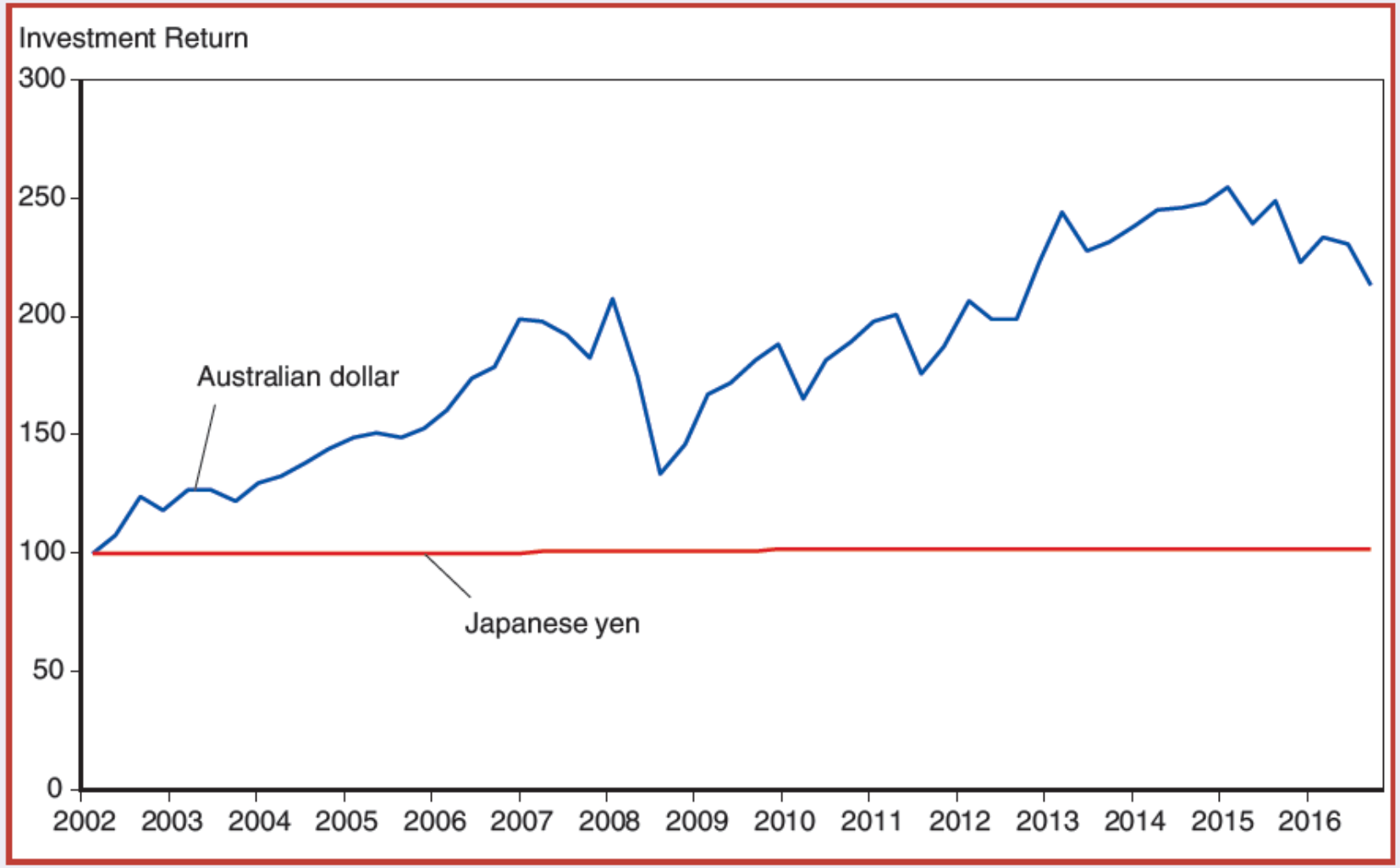


FIGURE 14-7

Cumulative Total Investment Return in Australian Dollar Compared to Japanese Yen, 2003–2016

The Australian dollar-yen carry trade has been profitable on average but is subject to sudden large reversals, as in 2008.

Source: Exchange rates and three-month treasury yields from Global Financial Data.

Carry Trade:
Borrow in Yen, invest in AUD

What can explain carry trade profits?

1. **A skewed distribution of exchange rate changes**
   - AUD appreciates most of the time but can abruptly depreciate:
     AUD goes up by the stairs and down by the elevator.
   - With prob. 90% AUD *appreciates* by 1% against the Yen
   - With prob. 10% AUD *depreciates* by 40% against the Yen
   - Expected rate of depreciation of Yen $= 0.9 \times 0.01 + 0.1 \times (-0.4) = -0.031$
   - **This can still be consistent with the UIP!** You need more time to see this (the Peso problem)

2. **Currency risk premium (when investors are risk averse)**
   - positive carry trade returns compensate for the above "disaster" risk (Brunnermeier, Nagel, Pedersen, 2008). That is, abrupt exchange rate movements occur during financial crises when other risky assets lose value [i.e., similar to equity risk premium]

Money and Interest Rates

What is Money?

- Lay English

  "That person has a lot of money"

  = "That person is wealthy"

- In Economics
  - "Money" has a more specific meaning
  - Roughly
    - The asset that people use to make transactions and set prices in terms of
  - Today: dollars, euros, yens, etc.
  - Past: gold, silver, corn, etc.

Money of Yap Island

Source: [Milton Friedman (1994), “Money Mischief”]

Money of Yap Island

- Yap coin (the rai) – stone wheel with a whole
- These huge stones were only used as money
- Coins were produced on Palau island
  (400 km away from Yap)
- Big enough to be hardly movable
- Yap monetary invention
  - They divorced ownership of the rai from physical control
  - Purchasing goods involved *public declaration* of the change in the ownership (everybody knows who owns a particular stone)
- One day a huge rai coin sank during transportation from Palau. This did not preclude it from later being used in transactions

Functions of Money

1. Store of Value: obvious

2. Medium of Exchange: money solves double-coincidence-of-wants problem

3. Unit of Account

Money Supply

How to measure money supply?

- **Idea**
  - Add things that people use for transactions

- **In reality**
  - Some assets are used more frequently in transactions than others
  - As a result, central banks compute several measures of money supply

Money Supply

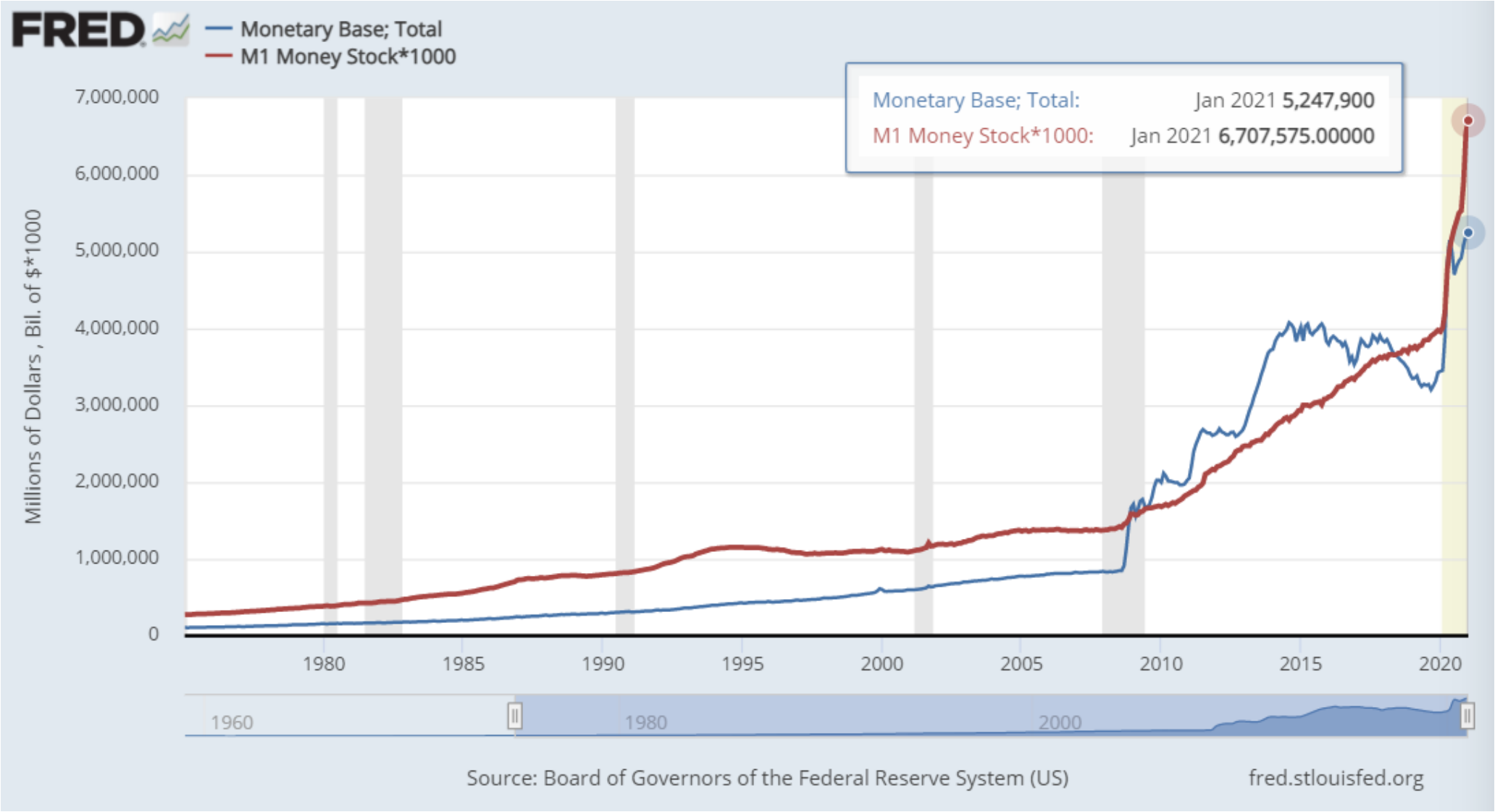
- Monetary base
  = notes and coins **in circulation**
  + notes and coins **in bank vaults**
  + *reserve deposits at the Federal Reserve*

  Bank reserves

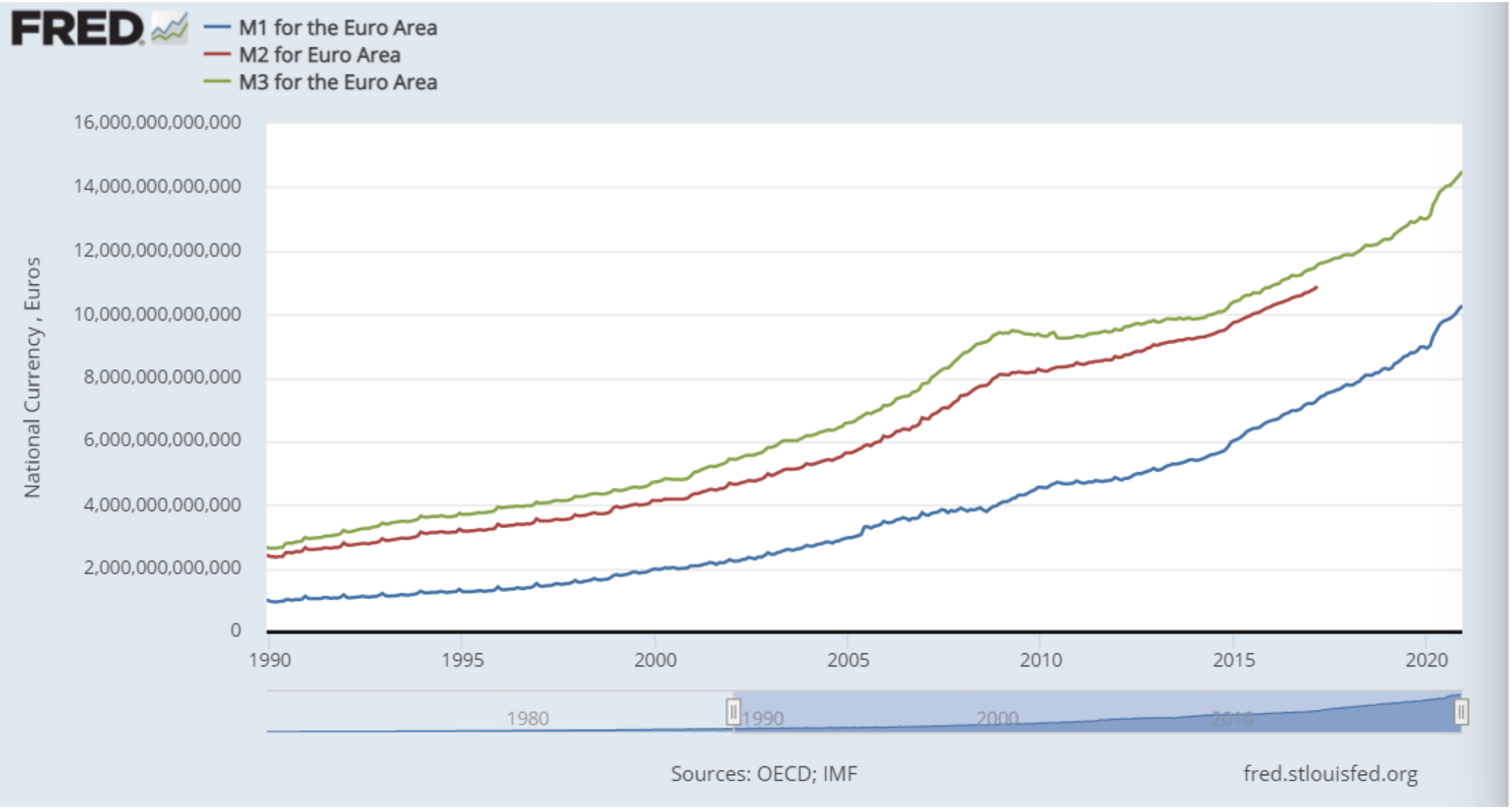
- $M1 =$ ~~*Monetary Base*~~
  notes and coins **in circulation**
  + demand deposits
  + other checkable deposits
- $M2 = M1$
  + retail money market mutual funds
  + savings and small time deposits
  + overnight repurchase agreements

- $M1 =$ currency, overnight deposits
- $M2 = M1$
  + deposits with agreed maturity up to 2 years
  + deposits redeemable at notice up to 3 months
- $M3 = M2$ + repurchase agreements, money market mutual funds debt securities up to 2 years' maturity

US Money Supply



Euro Area Money Supply



Money Demand

Individual Money Demand

- **Interest rates/expected rates of return**
  - (Usually) no interest on money
  - R = opportunity cost of holding money (interest rate on gov. bonds)
- **Risk**
  - the risk comes from unexpected inflation, which reduces the purchasing power of money.
  - But the same risk with other assets (e.g., bonds)!
- **Liquidity**
  - The fact that money is a convenient medium of transaction is the key reason why people want to hold it despite a lower return compared to government bonds
  - The volume of transactions will affect the demand for money

Aggregate Money Demands

- Aggregate money demand is the sum of individual money demands of people, financial and non-financial firms

- Determinants
  - Interest rates (opportunity cost)
  - Income
  - Price level (because money is usually a nominal variable)

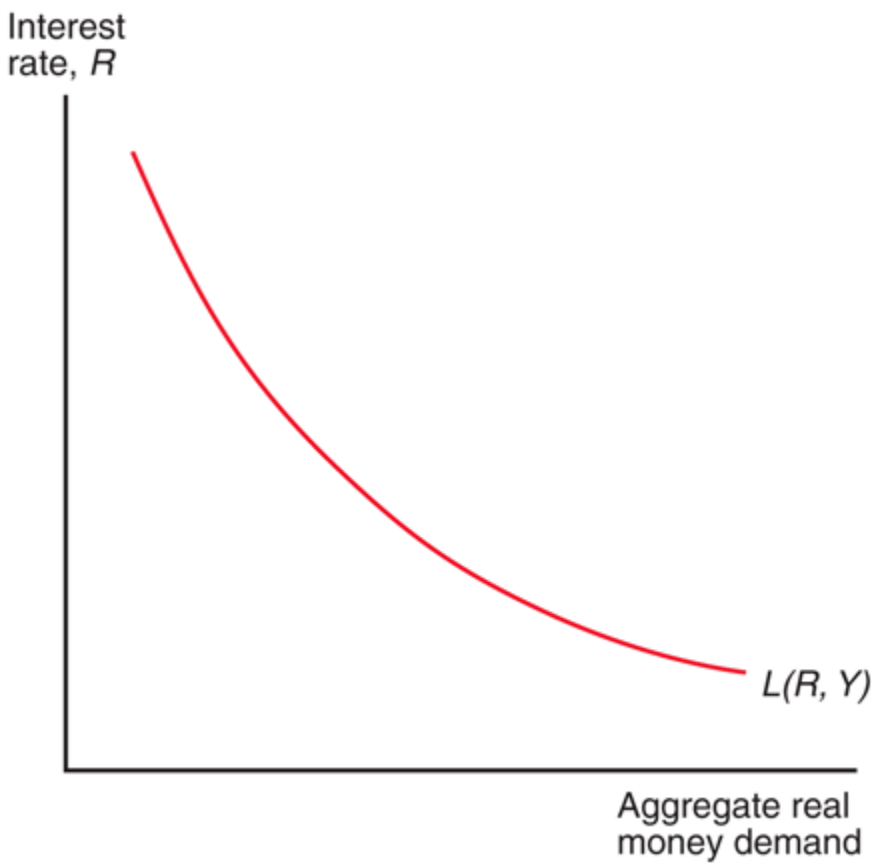
Aggregate Money Demand

- The aggregate demand for money can be expressed as:

$$\frac{M^d}{P} = L(R, Y)$$

  - P is the price level
  - Y is real national income
  - R is a measure of interest rates on nonmonetary assets
  - L(R,Y) is the aggregate real money demand

Aggregate Money Demand



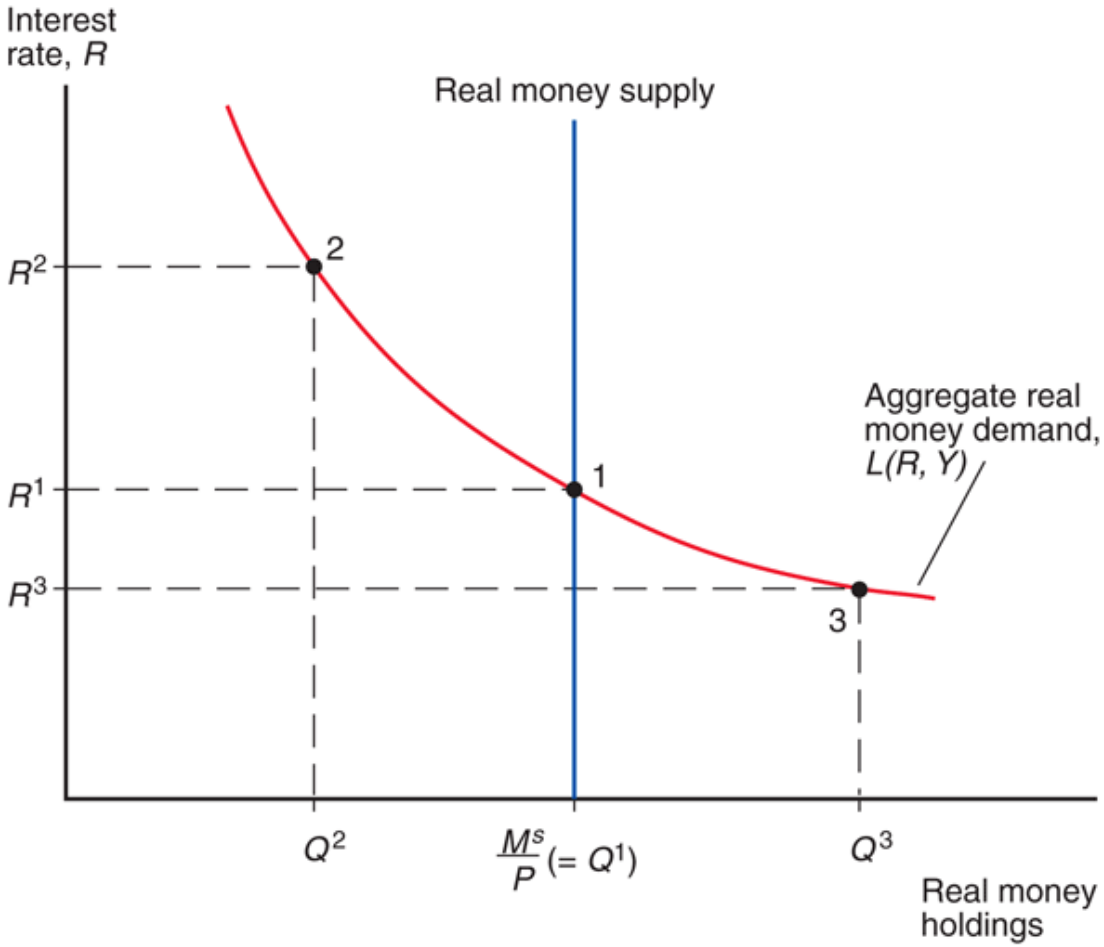
Money Market Equilibrium

Money Market Equilibrium

$$\frac{M^s}{P} = \frac{M^d}{P}$$

$$\frac{M^s}{P} = L(R, Y)$$

Money Market Equilibrium



Readings

- Next lecture:
  - KOM, Chapter 15

Money, Interest Rates, and Exchange Rates

Chapter 14 showed how the exchange rate between currencies depends on two factors—the interest that can be earned on deposits of those currencies and the expected future exchange rate. To understand fully the determination of exchange rates, however, we have to learn how interest rates themselves are determined and how expectations of future exchange rates are formed. In this and the next two chapters, we examine these topics by building an economic model that links exchange rates, interest rates, and other important macroeconomic variables such as the inflation rate and output.

The first step in building the model is to explain the effects of a country's money supply and of the demand for its money on its interest rate and exchange rate. Because exchange rates are the relative prices of national monies, factors that affect a country's money supply or demand are among the most powerful determinants of its currency's exchange rate against foreign currencies. It is therefore natural to begin a deeper study of exchange rate determination with a discussion of money supply and money demand.

Monetary developments influence the exchange rate by changing *both* interest rates *and* people's expectations about future exchange rates. Expectations about future exchange rates are closely connected with expectations about the future money prices of countries' products; these price movements, in turn, depend on changes in money supply and demand. In examining monetary influences on the exchange rate, we therefore look at how monetary factors influence output prices along with interest rates. Expectations of future exchange rates depend on many factors other than money, however, and these nonmonetary factors are taken up in the next chapter.

Once the theories and determinants of money supply and demand are laid out, we use them to examine how equilibrium interest rates are determined by the equality of money supply and money demand. Then we combine our model of interest rate determination with the interest parity condition to study the effects of monetary shifts on the exchange rate, given the prices of goods and services, the level of output, and market expectations about the future. Finally, we take a first look at the long-term effects of monetary changes on output prices and expected future exchange rates.

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